

DEPI – Software Testing Track

Part 01: Java Fundamentals

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Session 04

Assignment # 03

Java OOP Assignment – Shopping System in a Store

Imagine a simple store where customers buy products. A customer can add items to a shopping cart, and when the purchase is finished a bill is generated that shows all the items, their quantities, prices, and the total amount.

Your Task

Design a Java object-oriented solution using exactly the following four classes:

1. Item
2. Cart
3. Bill
4. Customer

For each class you must:

- List all necessary attributes (with their data types).
- List all necessary methods (constructors, getters, setters, and any other useful methods).
- Clearly indicate composition: state when one class is used as an attribute (or as an element of an ArrayList) inside another class.

Required Design Rules

- An Item represents a product that can be bought (name, price, etc.).
- A Cart belongs to a customer and contains a list of items (with quantities).
- A Bill is created from a cart and calculates the total amount to pay.
- A Customer has a name and can own one or more carts / bills.

Use ArrayList wherever a collection is needed.

Write a clear and complete class design that satisfies all the requirements above.